Verifier-Anchored RL over a Frozen LLM

Design, prior art, and first live results

Alexander Towell

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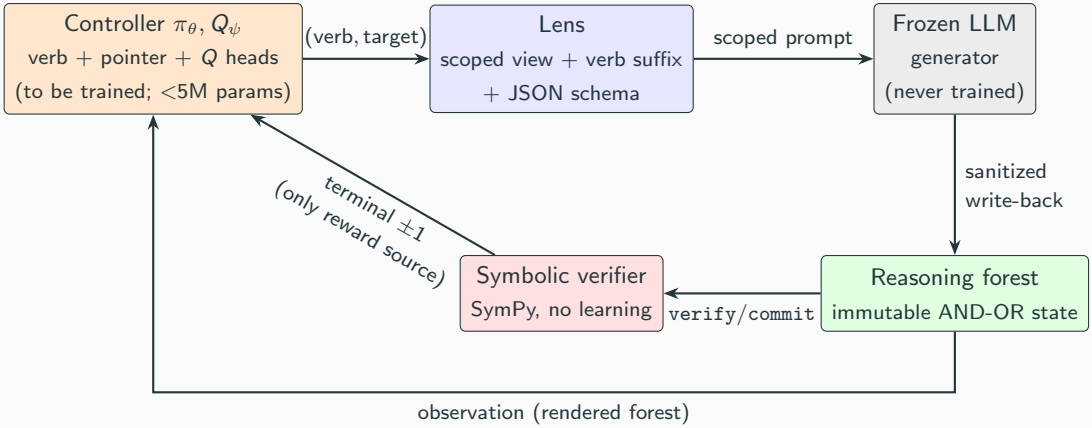
What this is (and is not)

The system

A tiny external controller (verb head + pointer head + Q head, a few million parameters) learns to orchestrate a frozen large language model (LLM) by issuing typed $(\text{verb}, \text{target})$ actions over a structured, nested reasoning forest. A symbolic verifier is the only reward source.

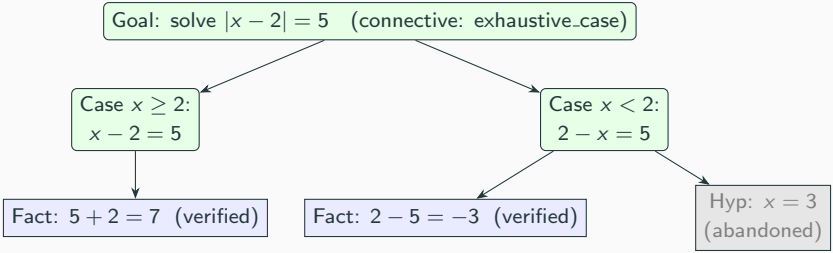
- **Not** o1 / DeepSeek-R1: reinforcement learning never touches the LLM's weights
- **Not** a learned reward model: no process or outcome reward model, no critic, no LLM-as-judge anywhere in the reward path
- Training spine: off-policy value-based Q-learning; Monte Carlo Tree Search (MCTS) is a strictly additive accelerator and the test-time compute knob

Architecture: one action selects the operation and scopes the context



`verify`/`commit`/`abandon` make no LLM call: the policy can neither author what is checked nor what it is checked against.

State: a nested AND-OR reasoning forest



- Frozen dataclasses: `Goal`, `Hypothesis`, `Fact`, `VerifierCall`; verbs are total functions Graph $\rightarrow$ Ok | Err
- No reward or value stored in nodes; `solved`/`verified` are derived predicates
- Structure never gates: the connective is metadata for the controller, not a hard rule
- Closed subtrees compact to one line in every rendered view (built-in roll-back / summarize)

Action space: six verbs, one pointer

Verb	LLM call?	Effect
`split`	yes	decompose a goal into subgoals or exhaustive cases
`form_hypothesis`	yes	add a tentative claim under a goal
`compute`	yes	add one checkable ground equation as a fact
`verify`	no	run the symbolic checker on a node's statement
`commit`	no	close a goal on verifier-vouched evidence
`abandon`	no	prune a dead end (excluded from menus and views)

The lens property

Choosing (verb, target) simultaneously picks the operation *and* the slice of state the frozen generator sees: problem + ancestor path + target subtree, siblings hidden, closed subtrees compacted. Write-back is local and sanitized against marker injection.

Reward invariant and the training ladder

Reward (load-bearing)

- Terminal $+1/-1$ at root `commit`; 0 elsewhere
- Masked/illegal action: small penalty
- No learned reward of any kind, ever
- Aesthetic signals may enter only as search priors, never as Bellman targets

Three rungs

1. LLM-as-controller: a strong teacher picks $(\text{verb}, \text{target})$; trajectories logged as JSON lines
2. Behavior cloning: warm-start π_θ on rung-1 data
3. Q-learning fine-tune: off-policy, n-step / Monte Carlo returns from the verifier; MCTS optional for better targets

Design-true collection regime: a strong *hosted* teacher model (reached over a network endpoint) decides; a *local small generator* writes all content, so trajectories reflect the deployed model's state distribution.

Prior art: the landscape

Cluster	Exemplars	How we differ
Search + train the LLM (distill / fine-tune)	TS-LLM, GSoS, OREO, Agent Q, rStar-Math	LLM frozen; controller is external and tiny
Learned process reward	ReST-MCTS*, rStar-Math, V-STaR, TS-LLM (ORM)	symbolic verifier is the *only* reward
Prompt-level RL over frozen LM	RLPrompt, TEMPERA	multi-step typed ops + pointer over a forest
External module + frozen LLM	PNLC (value only), UniR (logit level)	discrete action policy, operation-level Q
Context control (concurrent)	Active Context Curation (2026)	7B pruning policy vs $<$5M lens-as-byproduct

ORM: a learned *outcome* reward model (scores final answers); its process-level sibling scores intermediate steps. We forbid both.

Prior art: the three real threats, and what survives

Threats a reviewer will raise

1. TS-LLM read as "same loop, different reward source"
2. ArCHer (value RL that trains the LLM itself) + GSoS + verifiable-reward RL: "each ingredient is published, the combination is engineering"
3. Active Context Curation as concurrent work on the lens axis

What survives: the conjunction, claimed as such

- Tiny *external* controller (neither the LLM nor an LLM-sized critic)
- Factored verb + pointer heads over live node ids of a growing forest
- Off-policy value-based Q-learning as the spine, MCTS strictly additive
- Symbolic verifier as the sole reward, no learned value or reward model anywhere
- The lens property: action selection = context scoping, with local write-back

Honest caveat: defensible but thin per-element; claim the conjunction, not a new RL algorithm.

Casework: the structural action space fires

The dataset does its job (`casework-v1`: 300 SymPy-verified problems with provenance): on a linear benchmark the chain policy is optimal, so structural verbs never fire; on casework they reach 12–23% of decisions, with `split` in 20/20 probe episodes and `abandon` on the dead-case families.

The teacher-LLM controller plateaus at 20–25%. A win needs roughly 10 correct sequential decisions and per-step protocol fidelity binds; prompt engineering, a 5×-price teacher, longer completions, and coarser verified steps all measured null against pre-registered readings.

Why a trained controller can do better

The teacher's dominant errors are illegal action choices and repetition, errors a pointer head restricted to the legal menu cannot make by construction; and its logged decisions, right and wrong, are exactly the behavior-cloning data Phase B trains on.

The measured difficulty ladder, and the target

Direct prompt, same scoring	Qwen2.5-1.5B (deployment)	gpt-4.1-mini
quadratics (f3)	75%	100%
absolute value (f1a/f1b)	40 / 35%	100%
piecewise (f2a/f2b)	31 / 15%	100%
casework-v1 as shipped (the five above, 240)	39.2%	100%
held-out: word problems (f7) / counting (f8)	5% / 5%	98% / 100%

One generator spans 5–75% on the deployment model while staying trivial for the mini tier (harder mixes are a *weighting* exercise); dead-case problems are the small model's weak axis (33% vs 43%), exactly what `abandon` targets.

The decisive comparison, now concrete

Qwen2.5-1.5B + trained controller must beat 39.2% on casework-v1. Every teacher-side alternative (prompts, a 5×-price teacher, completion budget, solve-claim steps) measured null; the trained controller is the remaining lever.

Next steps

1. **Design-true collection**: hosted teacher decides, *local* Qwen-1.5B writes content. First probe 0/20: strict payload schemas mask 55–63% of the small model's typed-verb output; lenient coercion at that boundary is in progress
2. **Phase B**: per-node encoder + pointer head, behavior-cloning warm-start on rung-1 trajectories, then Q-learning fine-tune against the verifier
3. **The decisive comparison**: Qwen-1.5B + trained controller vs the same model prompted directly; the target is 39.2% on casework-v1, and the approach succeeds or fails on that margin
4. **Standard benchmarks later**: GSM8K/MATH follow once the casework loop shows progress

One-sentence summary

Search and scale live in the teacher and the compute knob; what we *learn* is a tiny verifier-anchored controller over a frozen LLM.